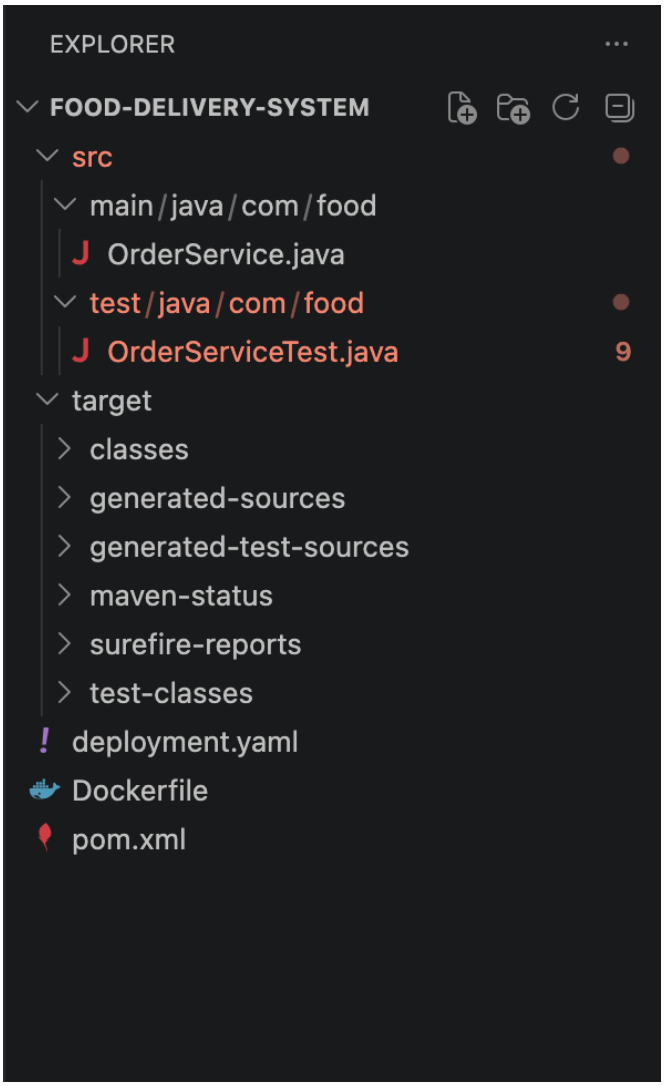
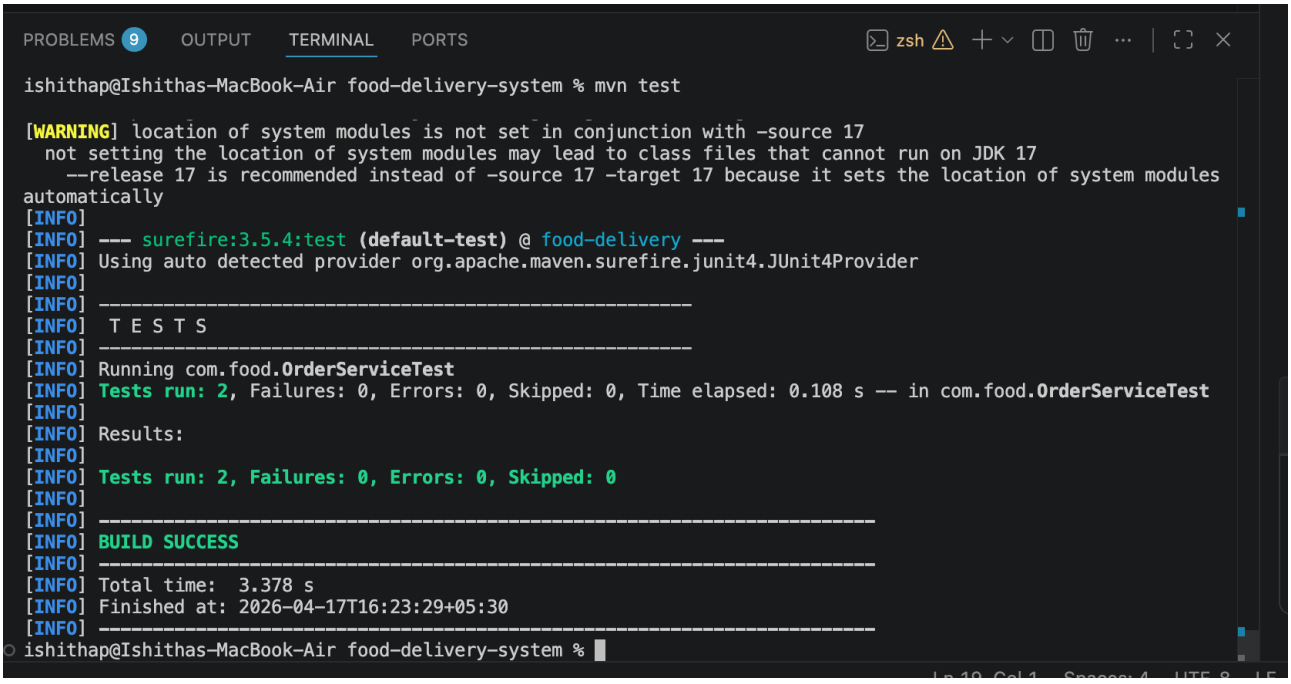
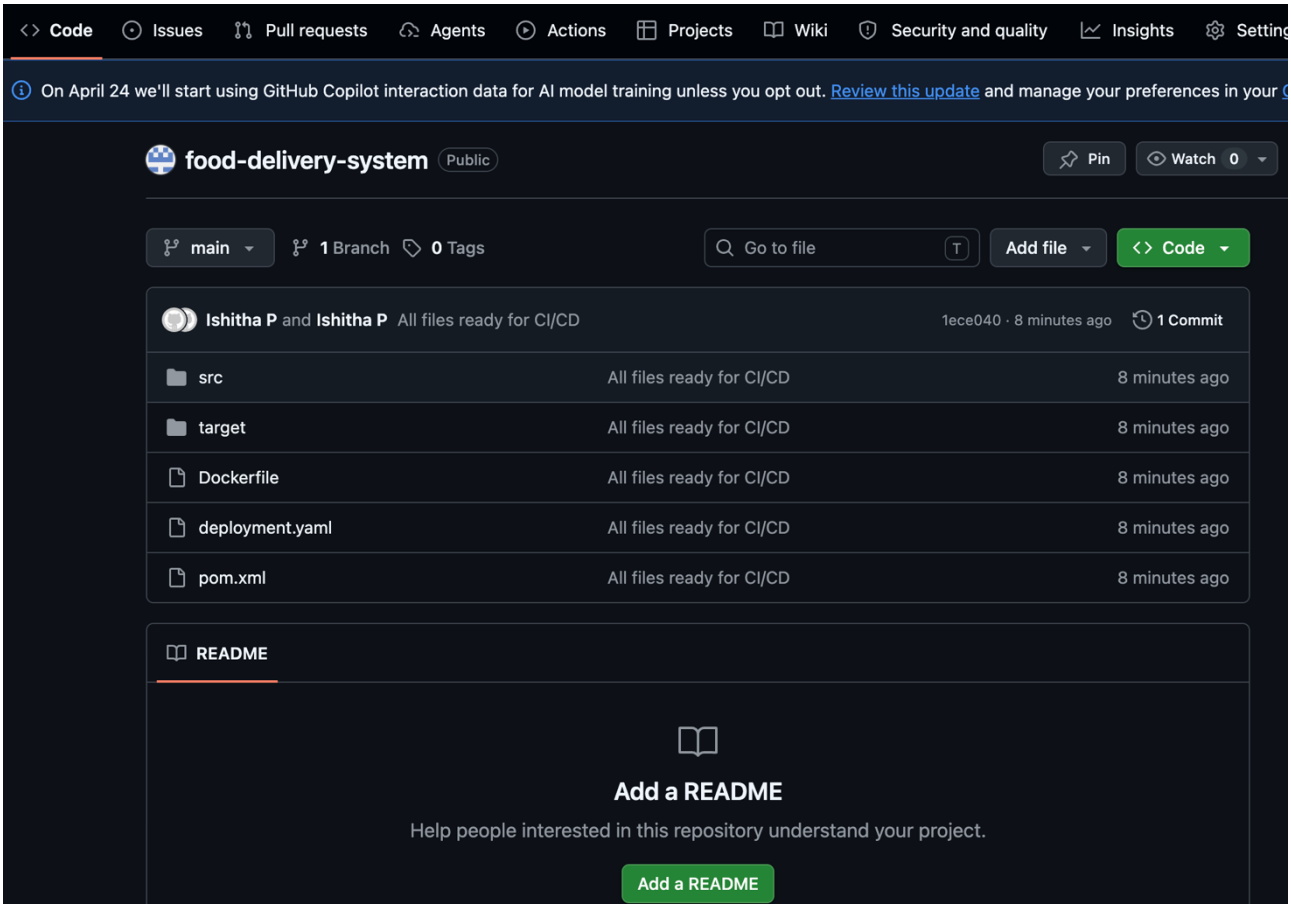
Java Project Structure and Order Management Service Logic in VS Code"



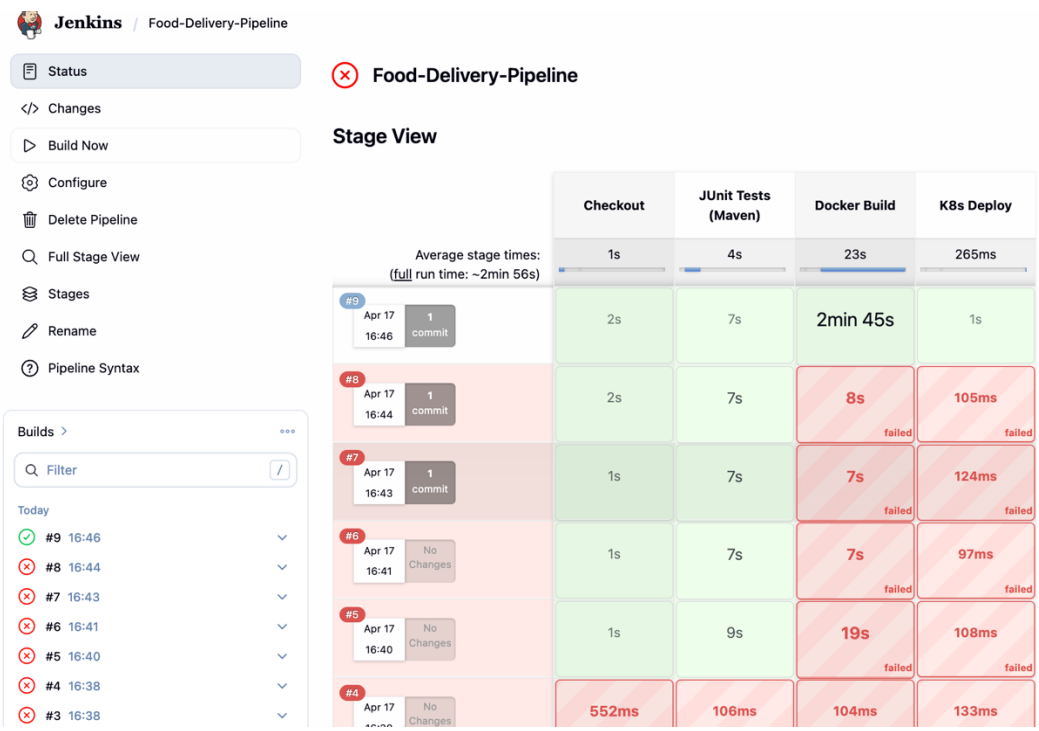
Successful Execution of JUnit Test Cases via Maven Lifecycle



Project Repository Hosted on GitHub for Version Control



Jenkins CI/CD Pipeline Stage View with Successful Automation



Verifying Application Deployment in Kubernetes Cluster

```
ishithap@Ishithas-MacBook-Air food-delivery-system % kubectl get pods
NAME                              READY   STATUS             RESTARTS   AGE
food-delivery-5878f55f69-wj5xg    0/1     ImagePullBackOff   0          15m
ishithap@Ishithas-MacBook-Air food-delivery-system %
```

Online Food Delivery System Container Image in Docker Desktop

food-delivery	latest	cdfa711f1e20	2 minutes ago	410.65 MB

Codes:

src/main/java/com/food/OrderService.java

```
package com.food;

public class OrderService {
    public String placeOrder(String item, double price) {
        if (price <= 0) {
            return "Order Failed";
        }
        return "Order Placed: " + item;
    }
}
```

src/test/java/com/food/OrderServiceTest.java

```
package com.food;

import static org.junit.Assert.*;
import org.junit.Test;

public class OrderServiceTest {
    @Test
    public void testOrderSuccess() {
        OrderService service = new OrderService();
        assertEquals("Order Placed: Pizza", service.placeOrder("Pizza", 10.0));
    }

    @Test
    public void testOrderFailure() {
        OrderService service = new OrderService();
        assertEquals("Order Failed", service.placeOrder("Burger", -5.0));
    }
}
```

pom.xml

```
<project xmlns="http://apache.org">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.food</groupId>
    <artifactId>food-delivery</artifactId>
    <version>1.0-SNAPSHOT</version>
```

```
    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
</project>
```

Dockerfile

```
FROM eclipse-temurin:17-jdk-jammy
COPY target/*.jar app.jar
ENTRYPOINT ["java","-jar","/app.jar"]
```

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: food-delivery
spec:
  replicas: 1
  selector:
    matchLabels:
      app: food
  template:
    metadata:
      labels:
        app: food
    spec:
      containers:
      - name: food-app
        image: food-delivery:latest
        ports:
        - containerPort: 8080
```

Pipeline script

```
pipeline {
    agent any
    environment {
        PATH = "/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin:/opt/homebrew/bin:$PATH"
        KUBECONFIG = "${HOME}/.kube/config"
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com'
            }
        }
        stage('JUnit Tests (Maven)') {
```

```
        steps {
            sh 'mvn clean test'
        }
    }
    stage('Docker Build') {
        steps {
            sh 'mvn package -DskipTests'
            sh 'docker build -t food-delivery:latest .'
        }
    }
    stage('K8s Deploy') {
        steps {
            sh 'kubectl --kubeconfig=${KUBECONFIG} apply -f deployment.yaml --validate=false'
        }
    }
}
}
```